

Vacuum Batch Distillation Ver 1

Pilot vacuum batch column C-101: methyl palmitate (C16:0) from palm-oil biodiesel

CAUSE AND EFFECT MATRIX

Cause and Effect Matrix · SIS-100 and interlocks of C-101

Document no.	VBD-IN-CE-001
Revision	R0
Safety system	SIS-100 (de-energise to trip)
Drawing reference	P&ID of the OTS (screen P&ID tab): C-101, D-101/D-102, HE-101 hot-oil loop, E-103, D-104, VP-101
Related documents	VBD-IN-SRS-001 Safety requirements specification (SRS of SIF tab); HAZOP and LOPA tabs of the OTS

REVISION RECORD

REV	DATE	DESCRIPTION	ORIGINATOR	CHECKER	APPROVER
R0	2026-10-09	Issued for teaching	Apinan Soottitantawat		

Teaching document generated from the Vacuum Batch Distillation Ver 1 operator training simulator. Trip settings, delays, effects and alarm limits are read from the simulator model (K.trip*, sisStep, sisTrip, ALM, armPermissives) and checked in the C&E matrix tab. Not for construction. Checker and approver fields are left blank on purpose.

CAUSE AND EFFECT MATRIX · VBD-IN-CE-001 · Vacuum Batch Distillation Ver 1

CAUSES							EFFECTS											NOTES	
ITEM	TAG	SERVICE	DIR.	SETTING	UNIT	DELAY s	TV-101 hot-oil valve: JIC-101 to MAN, output 0 %	HE-101 electric heater off: TIC-107 to MAN, output 0 %	XV-101 hot-oil shut-off valve closed (SIS not in RUN, fail closed)	TIC-101 on/off heating (as-built option) to MAN	XV-107 distillate diverter to CLOSED (total reflux)	SIS-100 state TRIPPED, reset required	First-out indication (SIS-100 faceplate)	DCS alarm in the alarm list	Audible horn	P-104 reflux pump stopped	Heating cannot be armed (permissive not met)	TRIP ACTION	
							CLOSE	OFF	CLOSE	MAN	CLOSED	TRIP	1ST	ALARM	HORN	STOP	BLOCK		
TRIPS OF SIS-100 (SIF-01 ... SIF-05) AND ESD																			
001	HS-100	Emergency stop push-button (DCS console / SIS-100 faceplate)	—	push		—	X	X	X	X	X	X	X	X	X			Operator ESD; first-out "HS-100 operator emergency stop"	
002	TSHH-101	Pot D-101 temperature, TE-101B (own sensor)	HH	290	°C	5	X	X	X	X	X	X	X	X	X			Active in every SIS state; reset blocked while above the setting	
003	PSHH-102	Column top pressure, PT-102 (shared with PIC-102)	HH	300	mbar	5	X	X	X	X	X	X	X	X	X			Active only in RUN	
004	FSLL-108	Tempered water to E-103, FT-108 (shared with FI-108)	LL	0.36	m³/h	10	X	X	X	X	X	X	X	X	X			Active only in RUN	
005	LSLL-101	Pot D-101 level, LT-101 (shared with LI-101)	LL	20	%	5	X	X	X	X	X	X	X	X	X			Active only in RUN	
006	TSHH-107	Hot-oil supply temperature, TE-107B (own sensor)	HH	330	°C	5	X	X	X	X	X	X	X	X	X			Active in every SIS state; reset blocked while above the setting	
EQUIPMENT INTERLOCK																			
007	LSLL-104	Reflux drum D-104 level (P-104 pump protection)	LL	5	%	5										X		Equipment interlock, not a SIF; LIC-104 L alarm at 20 % comes first	
DCS ALARMS (ALM LIMITS)																			
008	PIC-102	Column top pressure	L	140	mbar	—								X	X			Alarm; suppressed while VP-101 is stopped	
009	PIC-102	Column top pressure	H	200	mbar	—								X	X			Alarm; suppressed while VP-101 is stopped	
010	PDI-103	Column ΔP high (flooding)	H	18	mbar	—								X	X			Alarm; suppressed unless boiling in RUN	
011	TIC-101	Pot temperature high	H	280	°C	—								X	X			Alarm	
012	TIC-107	Hot-oil supply high	H	315	°C	—								X	X			Alarm	
013	JIC-101	Pot duty high (flooding risk)	H	5	kW	—								X	X			Alarm; suppressed unless boiling in RUN	
014	LIC-104	Reflux drum level	L	20	%	—								X	X			Alarm; suppressed while P-104 is stopped	
015	LIC-104	Reflux drum level	H	85	%	—								X	X			Alarm; suppressed while P-104 is stopped	
016	LI-101	Pot level low	L	30	%	—								X	X			Alarm; suppressed unless SIS-100 is in RUN	
017	FI-108	Tempered-water flow low	L	0.6	m³/h	—								X	X			Alarm; suppressed when SIS-100 is not in RUN and P-108 is stopped	
018	TI-109	Tempered-water temperature	L	45	°C	—								X	X			Alarm; suppressed when SIS-100 is not in RUN and P-108 is stopped	
019	TI-109	Tempered-water temperature	H	80	°C	—								X	X			Alarm; suppressed when SIS-100 is not in RUN and P-108 is stopped	
020	TI-105	Condensate cold (freezing risk)	L	45	°C	—								X	X			Alarm; suppressed unless boiling in RUN	
021	FIC-104	Reflux flow low	L	20	kg/h	—								X	X			Alarm; suppressed while P-104 is stopped	
ARMING PERMISSIVES (SEQUENCE)																			
022	PIC-102	Arming permissive: VP-101 running and PIC-102 below 250 mbar	P	—		—											X	Sequence: checked by sisCmd("arm") from IDLE	
023	FI-108	Arming permissive: Tempered water P-108 running, FI-108 ≥ 0.6 m³/h	P	—		—											X	Sequence: checked by sisCmd("arm") from IDLE	
024	LI-101	Arming permissive: Pot level LI-101 above 30 %	P	—		—											X	Sequence: checked by sisCmd("arm") from IDLE	
025	P-101	Arming permissive: Hot-oil pump P-101 running	P	—		—											X	Sequence: checked by sisCmd("arm") from IDLE	
026	TIC-107	Arming permissive: Hot oil TIC-107 below 320 °C	P	—		—											X	Sequence: checked by sisCmd("arm") from IDLE	
027	HV-110	Arming permissive: N₂ vacuum break HV-110 closed	P	—		—											X	Sequence: checked by sisCmd("arm") from IDLE	
028	SPARE																		
029	SPARE																		

NOTES	
1. SIS-100 is de-energise to trip. On any trip: JIC-101 and TIC-107 go to MAN at 0 % (TV-101 shut, HE-101 off), XV-101 closes (hot oil only flows in RUN), TIC-101 goes to MAN, XV-107 goes to CLOSED (total reflux). VP-101, P-108 and P-104 keep running: vacuum and condenser stay on while the pot cools.	4. Delays are the sisStep() on-delays; the response time used in the SRS adds sensor 0.5 s, logic 0.25 s and valve stroke 2 s (assumed).
2. PSHH-102, FSLL-108 and LSLL-101 are active only in RUN and share their transmitter with the BPCS (PT-102, FT-108, LT-101). TSHH-101 and TSHH-107 have their own elements and act in every state.	5. LSLL-104 is a pump-protection interlock in the equipment logic, not a SIF. Alarm rows carry the ALM limits of the DCS; the alarm deadband is 0.4 % of the alarm range (min 0.02).
3. Reset (SIS-100 faceplate) is refused while a non-RUN trip condition (TSHH-101, TSHH-107) is still present; after reset the state is IDLE and heating must be re-armed (permissive rows).	6. Rows 001 onward are numbered in the PDF; the two last rows are spare.

PROJECT		Pilot vacuum batch column C-101: methyl palmitate (C16:0) from palm-oil biodiesel			
CLIENT		Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University			
UNIT		Vacuum Batch Distillation Ver 1 · SIS-100 (de-energise to trip)			
DOCUMENT NO.		VBD-IN-CE-001			
TITLE		Cause and Effect Matrix · SIS-100 and interlocks of C-101			
P&ID		P&ID of the OTS (screen P&ID tab): C-101, D-101/D-102, HE-101 hot-oil loop, E-103, D-104, VP-101			
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